

ANIL PRADHAN

Full Stack Developer · AI/Agent Systems · Backend Engineering

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EDUCATION

B.Tech – CSE | CGPA: 9.11/10 | ITER, SOA University, Bhubaneswar 2024 – 2028
Class XII – PCM with IT | 85% | Odisha Board of Secondary Education 2022 – 2024

EXPERIENCE

Campus Ambassador | E-Cell, IIT Kanpur · Remote Dec 2025 – Jan 2026

- Represented IIT Kanpur's E-Summit at ITER, SOA University; boosted student participation via tech talks and networking sessions.

PROJECTS

AI Career Mentor

Feb 2026 – Present

[Next.js 14](#) · [FastAPI](#) · [Microsoft AutoGen v0.2](#) · [Neon Postgres](#) · [Groq](#) · [Gemini 2.5 Flash](#) · [WebSocket](#) · [Upstash Redis](#) · [Google OAuth](#) · [Tavily AI](#) · [Serper.dev](#)

- Architected a production-grade Multi-Agent AI platform using Microsoft AutoGen v0.2, implementing a GroupChatManager with custom speaker selection to orchestrate 4 specialized agents (Resume Analyst, Market Researcher, Career Coach, LinkedIn Reviewer) via `asyncio.to_thread`, delivering complex career diagnostics in <60 seconds.
- Engineered a high-performance WebSocket mock interview system using the OpenAI SDK integrated with Groq (Llama 3.3 70B), achieving sub-2-second token streaming latency with a 7-question adaptive state machine that dynamically scales technical difficulty based on company tiers (FAANG to Service-based).
- Implemented Hybrid Dual-LLM Architecture with automatic programmatic failover — Gemini 2.5 Flash as primary reasoning engine with seamless fallback to Groq/Llama-3 on 429 rate limit errors, ensuring 100% system availability and zero user-facing failures.
- Hardened Market Intelligence Pipeline by integrating Tavily AI & Serper.dev with regex-based JSON extraction, fetching real-time salary benchmarks, hiring volumes, and trending skills from live Google Search data — eliminating all static mock dependencies.
- Built Distributed Rate Limiting & AI Caching using Upstash Redis and SlowAPI middleware (100 req/hr, 1000 req/day per user); SHA-256 keyed response caching to minimize API costs, with usage counters incrementing only on successful AI generation.
- Designed Secure Auth & Data Pipeline — Google OAuth 2.0 (ID Token verification via google-auth), short-lived JWTs with Refresh Tokens, deterministic ATS Scoring Engine, and in-memory PDF processing via pdfplumber for structured JSON storage in Neon Serverless Postgres.
- Deployed & Automated Microservices across Vercel (Next.js 14) and Render (FastAPI) via GitHub Actions CI/CD with integrated ESLint, pytest, and pip-audit security scanning for production-grade reliability.

VolunteerIQ

Mar 2026 – Apr 2026

[Next.js 14](#) · [FastAPI](#) · [Firebase Auth](#) · [Gemini 2.5 Flash](#) · [Mapbox GL](#) · [PostgreSQL](#) · [Docker](#)

- Built an AI-powered NGO coordination platform for Google Solution Challenge 2026, integrating Gemini 2.5 Flash to process multi-format field surveys (PDF, CSV, DOCX, TXT) and extract urgent community problems with urgency scores and auto-generated deployable field tasks.
- Implemented AI volunteer matching system ranking candidates by skill match, location proximity, and availability with a 0–100 score, alongside a context-aware AI chat assistant querying live PostgreSQL data to answer plain-English NGO operational queries.
- Deployed a Mapbox GL live operations map with geocoding cache, Firebase JWT authentication verified server-side on every protected request, Docker + docker-compose containerization, and Gemini-to-Groq automatic fallback for uninterrupted AI service under rate limits.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · Java

Frontend: Next.js 14 · React · Vanilla CSS · Recharts · Lucide React · Axios · react-map-gl

Backend: FastAPI · Microsoft AutoGen v0.2 · OpenAI SDK (via Groq) · SlowAPI · SQLAlchemy · Alembic · Pydantic · Uvicorn · Loguru · pdfplumber · edge-tts

Databases: Neon Postgres · SQLite · Upstash Redis

AI / LLM: Google Gemini 1.5 Flash · Groq (Llama 3.3 70B) · Microsoft AutoGen

Auth & Security: Google OAuth 2.0 · Firebase Authentication · JWT · bcrypt · google-auth

DevOps & Infra: Docker · GitHub Actions · Vercel · Render.com · CI/CD

Tools: VS Code · Git · Swagger UI · Postman · Mapbox GL

ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Skills Boost Arcade Program: Earned 75+ points and multiple rewards through hands-on cloud labs and challenges focused on Google Cloud technologies.
- Major International Hackathons: Participated in Microsoft AI Dev Days Hackathon, Amazon Nova AI Challenge, and Google Solution Challenge 2026 — working continuously for multiple months on AI-based project development and problem-solving.
- Competitive Programming: Solved 200+ problems across LeetCode and GeeksforGeeks; earned a certificate and T-shirt for successfully completing the GeeksforGeeks 80 Days Coding Challenge.
- Active Tech Community: Participated in multiple offline technical workshops, coding competitions, and developer events including Google Developer Groups (GDG) sessions — improving practical knowledge in software development, AI, and modern technologies.